



Habib University

Course Code: EE252

Course Title: Signals and Systems – Spring 2025

Instructor's Name: Tariq Mumtaz

Examination: Quiz 1

Date: 6th Feb, 2025

Total Marks: 50 (5% Course credit)

Duration: 50 minutes

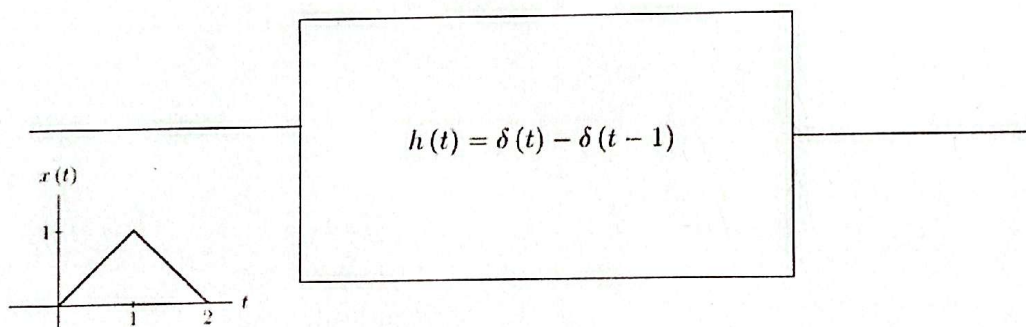
CLO: 01; CDL: 3, PLO: 1

Q1[20]. Demonstrate that the convolution of two square pulses with different widths produces a trapezoidal waveform. Provide a detailed step-by-step solution, including mathematical expressions and graphical plots for each stage of the convolution process.

$$x(t) = \Pi\left(\frac{t-2}{4}\right), \quad h(t) = \Pi\left(\frac{t-3}{6}\right)$$

Q2[10] By direct application of the convolution property below, determine the expression and sketch the output $y(t)$ of the given LTI system

$$x(t) * \delta(t - t_0) = x(t - t_0)$$



Q3[20] Determine which properties hold for each of the following continuous-time systems, a differentiator and an integrator. Justify your answers by applying appropriate property/law.

- Memoryless
- Time-Invariant
- Linear

$$y(t) = \frac{dx(t)}{dt}$$

$$y(t) = \int_{-\infty}^t x(\lambda) d\lambda$$

$$y(t-t_0) = \int_{-\infty}^{t-t_0} x(\lambda-t_0) d\lambda$$

$$\int_{-\infty}^t x(\lambda-t_0) d\lambda$$

$\lambda - t_0 = z$

$$\int_{-\infty}^{t-t_0} x(z) dz = \int_{-\infty}^{t-t_0} x(z) dz$$

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Q. 3) (i) $y(t) = \frac{d}{dt} x(t)$ $y_1(t) = \frac{d}{dt} x_1(t)$ $y_2(t) = \frac{d}{dt} x_2(t)$

$$\rightarrow y(t) = \frac{d}{dt} \{ \alpha_1 x_1(t) + \alpha_2 x_2(t) \}$$

$$y(t) = \alpha_1 \frac{d}{dt} x_1(t) + \alpha_2 \frac{d}{dt} x_2(t) \quad (\text{Linear})$$

$$y(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$$

$$\rightarrow \cancel{x(t)} = x(t-t_0) \rightarrow y(t-t_0) = \frac{d}{dt} x(t-t_0)$$

$$y(t) = \frac{d}{dt} x(t) \rightarrow y(t-t_0) = \frac{d}{dt} x(t-t_0)$$

(Time Invariant)

$\rightarrow$ Since output only depends on current value of input its Memoryless.

so for $y(t) = \frac{d}{dt} x(t)$: Memoryless (✓)

Time-Invariant (✓)

Linear (✓) by law of superposition.

$$(ii) y(t) = \int_{-\infty}^t x(\lambda) \cdot d\lambda \quad y_1(t) = \int_{-\infty}^t x_1(\lambda) \cdot d\lambda, y_2(t) = \int_{-\infty}^t x_2(\lambda) \cdot d\lambda$$

$$\rightarrow y(t) = \int_{-\infty}^t \{ \alpha_1 x_1(\lambda) + \alpha_2 x_2(\lambda) \} \cdot d\lambda$$

$$= \alpha_1 \int_{-\infty}^t x_1(\lambda) \cdot d\lambda + \alpha_2 \int_{-\infty}^t x_2(\lambda) \cdot d\lambda$$

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$$y(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$$

by Law of superposition it's linear.

$$\rightarrow x(\lambda) = x(\lambda - t_0) \rightarrow y(t - t_0) = \int_{-\infty}^t x(\lambda - t_0) \cdot d\lambda$$

$$y(t) = \int_{-\infty}^t x(\lambda) \cdot d\lambda \rightarrow y(t - t_0) = \int_{-\infty}^{t - t_0} x(\lambda) \cdot d\lambda$$

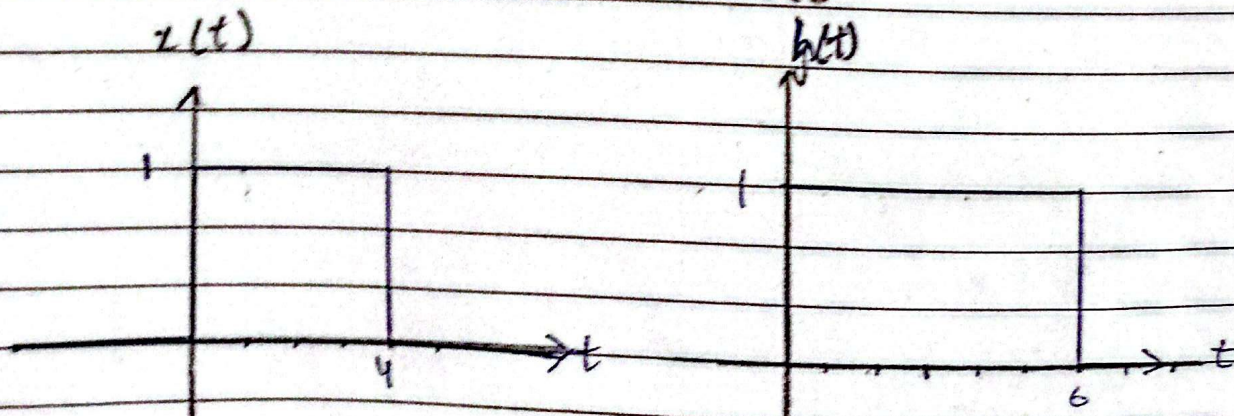
(Not time-invariant)

$\rightarrow$ It also depends on its past value so it does have memory.

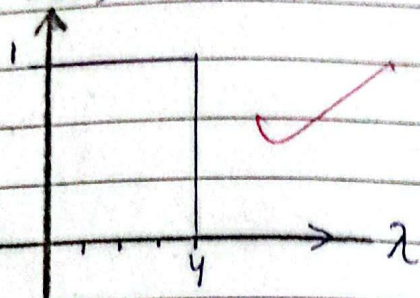
so for $y(t) = \int_{-\infty}^t x(\lambda) \cdot d\lambda$: Memoryless (X)
Time Invariant (X)
Linear (✓)

$$Q.1) x(t) = \Pi\left(\frac{t-2}{4}\right) = \begin{cases} 1 & 0 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

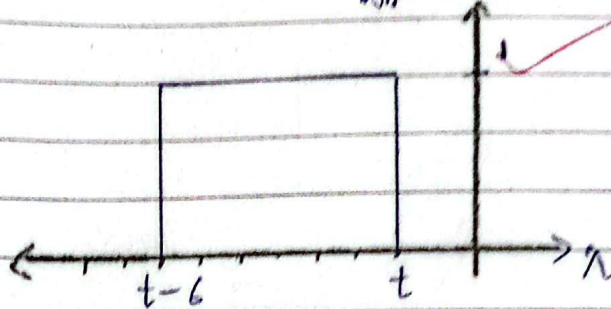
$$h(t) = \Pi\left(\frac{t-5}{6}\right) = \begin{cases} 1 & 0 < t < 6 \\ 0 & \text{otherwise} \end{cases}$$



$x(\lambda)$

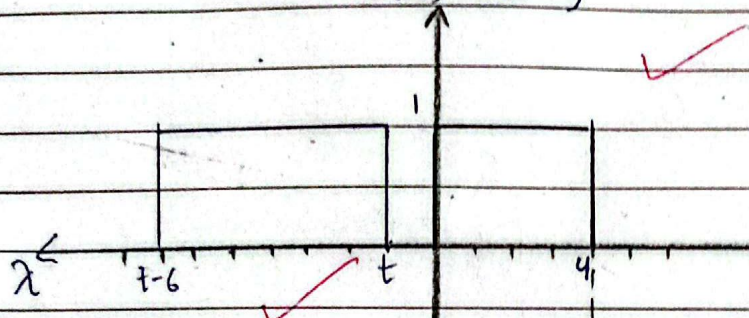


$h(t-\lambda)$



Case # 1

$x(\lambda) * h(t-\lambda)$



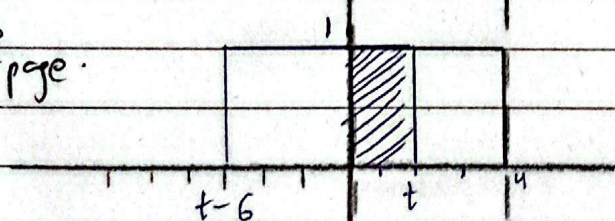
for $t \leq 0$
there is no overlapping
here
 $\int_{-\infty}^0 x(\lambda) h(t-\lambda) \cdot d\lambda = 0$

$$A = 0$$

Case # 2

$x(\lambda) * h(t-\lambda)$

* Case 3
on next page.



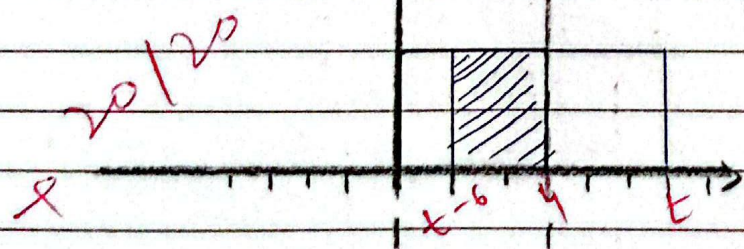
for $0 < t < 4$

$$= \int_0^t x(\lambda) h(t-\lambda) \cdot d\lambda$$

$$= [\lambda]_0^t = t$$

Case # 4

$x(\lambda) * h(t-\lambda)$



for $4 < t < 10$

$$= \int_{t-6}^4 x(\lambda) h(t-\lambda) \cdot d\lambda$$

$$= [\lambda]_{t-6}^4$$

$$= 4 - t + 6$$

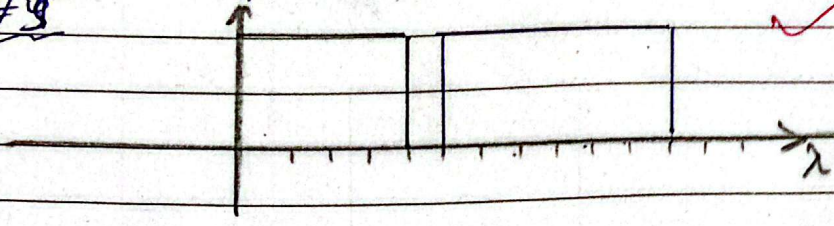
$$= 10 - t$$

one region is
missing
 $4 < t < 6$
 $4 < t < 10$

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Case #5

$x(\lambda) * h(t-\lambda)$



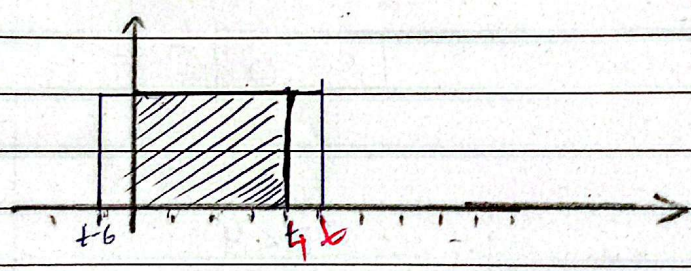
for $t > 10$

$$\int_{t-6}^{\infty} x(\lambda) h(t-\lambda) \cdot d\lambda = 0$$

$$x(\lambda) * h(t-\lambda) = \begin{cases} 0 & , t < 0 \\ t & , 0 < t < 4 \\ 10-t & , 4 < t < 10 \\ 0 & , t > 10 \\ 4 & , 4 < t < 6 \end{cases}$$

Case #3

$x(\lambda) h(t-\lambda)$



for $4 < t < 6$

$$= \int_0^4 x(\lambda) h(t-\lambda) \cdot d\lambda = [1]_0^4 = 4$$

~~$y(t) = x(t) * h(t)$~~
 ~~$y(t) = x(t) * h(t)$~~

